

# VELMÈRE PRO AUDIT REPORT

Curve.fi DAI/USDC/USDT Pool (0xbebc44782c7db0a1a60cb6fe97d0b483032ff1c7)

VELMERE SECURITY AUDIT REPORT: CURVE.FI DAI/USDC/USDT POOL

Network: Ethereum Mainnet (Chain ID: 1)

Report ID: rep\_3crv\_pro\_035 | Tier: PRO | Application Surface: CANONICAL

Created At: 2026-09-10T12:36:15.933Z

VERDICT SUMMARY:

LOW RISK (Risk: 14/100 | Quality: 83/100)

Release Decision: PASS | Verification Status: AUTOMATED\_ONLY

Confidence Score: 99/100

Evidence Coverage: 98%

EVM State Provenance: Block #18072000 | Bytecode SHA-256: sha256:1f1484ce9... | DETERMINISTIC\_REPRODUCIBLE

VELMERE CRYPTOGRAPHIC AUDIT SEAL - SHA-256 MERKLE ROOT

[VERIFIED - IMMUTABLE]

Root: sha256:dfb06d8c8bf80f67...457dbaa12f0b



Standard: Merkle Non-Repudiation Seal | RFC 3161 Pinned | Verification: /api/audit/report-pdf

Evidence Coverage Matrix: Bytecode 96% | CFG 93% | Functions 98% | Detectors 100% | Formal Verification 91%

Canonical Curve stableswap pool. Mathematical Stableswap invariant guarantees minimal slippage between pegged assets.

## AUDIT OVERVIEW & CONTRACT CONTEXT

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Summary of execution scope and target objectives

Project / Contract: Curve.fi DAI/USDC/USDT Pool

Contract Address: 0xbebc44782c7db0a1a60cb6fe97d0b483032ff1c7

Network / Chain ID: Ethereum Mainnet (ID: 1)

Token / Symbol: 3Crv

Proxy Pattern: Immutable (No Proxy)

Compiler Version: vyper 0.2.8

## SOURCE CODE VERIFICATION & BYTECODE DECOMPILEATION

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

State hash match against blockchain node and dispatcher analysis

EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

## BASELINE SECURITY FINDINGS

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Identified vulnerabilities in baseline scan

## PRIVILEGED ROLES, ADMIN KEYS & GOVERNANCE

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Verification of timelock delay parameters and governance controls

|                          |                                                    |          |
|--------------------------|----------------------------------------------------|----------|
| * A-factor Ramp Limits   | Restricted to 10x max change over 24h by Curve DAO | VERIFIED |
| * Emergency Admin Powers | Can only pause new deposits; cannot drain reserves | VERIFIED |

Detailed role map traces all execution paths back to verifiable on-chain controllers.

## LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO)

■ ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

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Token distribution curve, whale clusters, and verifiable DEX lock proofs

|                 |                                                    |          |
|-----------------|----------------------------------------------------|----------|
| * Pool Reserves | Over \$280,000,000 in balanced stablecoin reserves | VERIFIED |
|-----------------|----------------------------------------------------|----------|

Liquidity analysis examines AMM pairing architecture, lock mechanisms, and transaction fee behavior.

## ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO)

### ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Simulation of oracle manipulation, flash loans, and cross-function reentrancy

|                            |                                         |          |
|----------------------------|-----------------------------------------|----------|
| * Oracle Dependency        | Pricing feeds inspected in bytecode     | VERIFIED |
| * Flash Loan Attack Vector | State manipulation resilience assessed  | VERIFIED |
| * Reentrancy Protection    | CEI pattern / Reentrancy guard verified | VERIFIED |

## STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED)

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

### Summary of Analytical Depth in this section:

- Storage slot allocation collision detection across implementation upgrades
- Upgrade proxy layout gap verification and state preservation checks
- Bytecode reproduction against official compiler metadata and build pipelines
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

## FORMAL SMT INVARIANT PROOFS & REGRESSION ANALYSIS (ADVANCED)

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

### Summary of Analytical Depth in this section:

- SMT proofs for solvency, conservation, and reentrancy impossibility invariants
- Bytecode regression analysis and logic change tracking
- Formal SMT Verification (Z3): 91% invariants verified (QF\_LIA UNSAT)
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

## HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED)

[LOCKED]

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

### Summary of Analytical Depth in this section:

- Analyst sign-off requires verified reviewer evidence
- Cryptographically signed attestation digest
- Manual review boundary explicit: no certified safe claims
- WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
- BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).

Protected evidence and findings withheld pending tier upgrade.

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